
Algorithm 4 Pushforward Sampling

Input: model f^θ , time steps $\{t_i\}_{i=0}^N$, prior distribution $\mathcal{N}(0, \sigma_d^2 I)$, (optional) guidance weight w
Output: $\mathbf{x}_{t_0}$
Sample $\mathbf{x}_N \sim \mathcal{N}(0, \sigma_d^2 I)$
for $i = N, \dots, 1$ **do**
 (Optional) $w \leftarrow 1$ if $N = 1$ // can optionally discard unconditional branch for $N = 1$
 $\mathbf{x}_{t_{i-1}} \leftarrow f_{t_{i-1}, t_i}^\theta(\mathbf{x}_{t_i})$ or $f_{t_{i-1}, t_i, w}^\theta(\mathbf{x}_{t_i})$
end for

Algorithm 5 Restart Sampling

Input: model f^θ , time steps $\{t_i\}_{i=0}^N$, prior distribution $\mathcal{N}(0, \sigma_d^2 I)$, DDIM interpolant coefficients α_t and σ_t , (optional) guidance weight w
Output: $\mathbf{x}_{t_0}$
Sample $\mathbf{x}_N \sim \mathcal{N}(0, \sigma_d^2 I)$
for $i = N, \dots, 1$ **do**
 (Optional) $w \leftarrow 1$ if $N = 1$ // can optionally discard unconditional branch for $N = 1$
 $\tilde{\mathbf{x}} \leftarrow f_{t_0, t_i}^\theta(\mathbf{x}_{t_i})$ or $f_{t_0, t_i, w}^\theta(\mathbf{x}_{t_i})$
 if $i \neq 1$ **then**
 $\tilde{\epsilon} \sim \mathcal{N}(0, \sigma_d^2 I)$
 $\mathbf{x}_{t_{i-1}} \leftarrow \alpha_{t_{i-1}} \tilde{\mathbf{x}} + \sigma_{t_{i-1}} \tilde{\epsilon}$ // or more generally $\mathbf{x}_{t_{i-1}} \sim q_t(\mathbf{x}_{t_{i-1}} | \tilde{\mathbf{x}}, \tilde{\epsilon})$
 else
 $\mathbf{x}_{t_0} = \tilde{\mathbf{x}}$
 end if
end for

where $w(s, t)$ is a weighting function. If s is a small positive constant, we further have $f_{s, t}^\theta(\mathbf{x}_t) \approx g_\theta(\mathbf{x}_t, t)$ where we drop s as input. If $g_\theta(\mathbf{x}_t, t)$ itself satisfies boundary condition at $s = 0$, we can directly take $s = 0$ in which case $f_{0, t}^\theta(\mathbf{x}_t) = g_\theta(\mathbf{x}_t, t)$. And under these assumptions, our loss becomes

$$\mathbb{E}_{\mathbf{x}_t, \mathbf{x}, t} \left[w(t) \|g_\theta(\mathbf{x}_t, t) - g_\theta(\mathbf{x}_r, r)\|^2 \right], \quad (88)$$

which is simply a CM loss using ℓ_2 distance. □

However, one can notice that from a moment-matching perspective, this loss significantly deviates from a proper divergence between distributions, and is problematic in two aspects. First, it assumes single-particle estimate, which now ignores the entropy repulsion term in MMD that arises only during multi-particle estimation. This can contribute to mode collapse and training instability of CM. Second, the choice of energy kernel is not a proper positive definite kernel required by MMD. At best, it only matches the *first* moment (its Taylor expansion cannot cover all moments as in RBF kernels), which is insufficient for matching two complex distributions! We should use kernels that match higher moments in practice. In fact, we show in the following Lemma that the pseudo-huber loss proposed in [Song & Dhariwal \(2023\)](#) matches higher moments as a kernel.

Lemma 2. *Negative pseudo-huber loss $k_c(x, y) = c - \sqrt{\|x - y\|^2 + c^2}$ for $c > 0$ is a conditionally positive definite kernel that matches all moments of x and y where weights on higher moments depend on c .*

Proof. We first check that negative pseudo-huber loss $c - \sqrt{\|x - y\|^2 + c^2}$ is a *conditionally positive definite kernel* ([Auffray & Barbillon, 2009](#)). By definition, $k(x, y)$ is conditionally positive definite if for $x_1, \dots, x_n \in \mathbb{R}^D$ and $c_1, \dots, c_n \in \mathbb{R}^D$ with $\sum_{i=1}^n c_i = 0$

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j k(x_i, x_j) \geq 0 \quad (89)$$

We know that negative L_2 distance $-\|x - y\|$ is conditionally positive definite. We prove this below for completion. Due to triangle inequality, $-\|x - y\| \geq -\|x\| - \|y\|$. Then

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j (-\|x_i - x_j\|) \geq \sum_{i=1}^n \sum_{j=1}^n c_i c_j (-\|x_i\| - \|x_j\|) \quad (90)$$

$$= -\sum_{i=1}^n c_i \left(\sum_{j=1}^n c_j \|x_j\| \right) - \sum_{j=1}^n c_j \left(\sum_{i=1}^n c_i \|x_i\| \right) \quad (91)$$

$$\stackrel{(a)}{=} 0 \quad (92)$$

where (a) is due to $\sum_{i=1}^n c_i = 0$. Now since $c - \sqrt{\|z\|^2 + c^2} \geq -\|z\|$ for all $c > 0$, we have

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j \left(c - \sqrt{\|x_i - x_j\|^2 + c^2} \right) \geq \sum_{i=1}^n \sum_{j=1}^n c_i c_j (-\|x_i - x_j\|) \geq 0 \quad (93)$$

So negative pseudo-huber loss is a valid conditionally positive definite kernel.

Next, we analyze pseudo-huber loss's effect on higher-order moments by directly Taylor expanding $\sqrt{\|z\|^2 + c^2} - c$ at $z = 0$

$$\sqrt{\|z\|^2 + c^2} - c = \frac{1}{2c} \|z\|^2 - \frac{1}{8c^3} \|z\|^4 + \frac{1}{16c^5} \|z\|^6 - \frac{5}{128c^7} \|z\|^8 + \mathcal{O}(\|z\|^9) \quad (94)$$

$$= \frac{1}{2c} \|x - y\|^2 - \frac{1}{8c^3} \|x - y\|^4 + \frac{1}{16c^5} \|x - y\|^6 - \frac{5}{128c^7} \|x - y\|^8 + \mathcal{O}(\|x - y\|^9) \quad (95)$$

$$(96)$$

where we substitute $z = x - y$. Each higher order $\|x - y\|^k$ for $k > 2$ expands to a polynomial containing up to k -th moments, *i.e.*, $\{x, x^2, \dots, x^k\}, \{y, y^2, \dots, y^k\}$, thus the implicit feature map contains all higher moments where c contributes to the weightings in front of each term. $\square$

Furthermore, we extend our finite difference (between $r(s, t)$ and t) IMM objective to the differential limit by taking $r(s, t) \rightarrow t$ in Appendix H. This results in a new objective that similarly subsumes continuous-time CM (Song et al., 2023; Lu & Song, 2024) as a single-particle special case.

G.2. Diffusion GAN and Adversarial Consistency Distillation

Diffusion GAN (Xiao et al., 2021) parameterizes its generative distribution as

$$p_s^\theta(\mathbf{x}_s | \mathbf{x}_t) = \int q_{s|t}(\mathbf{x}_s | \mathbf{G}_\theta(\mathbf{x}_t, \mathbf{z}), \mathbf{x}_t) p(\mathbf{z}) d\mathbf{z}$$

where $\mathbf{G}_\theta$ is a neural network, $p(\mathbf{z})$ is standard Gaussian distribution, and $q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t)$ is the DDPM posterior

$$\begin{aligned} q_{s|t}(\mathbf{x}_s | \mathbf{x}, \mathbf{x}_t) &= \mathcal{N}(\mu_{s,t}, \sigma_{s,t}^2 I) \\ \mu_Q &= \frac{\alpha_t \sigma_s^2}{\alpha_s \sigma_t^2} \mathbf{x}_t + \alpha_s (1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}) \mathbf{x} \\ \sigma_Q^2 &= \sigma_s^2 (1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}) \end{aligned} \quad (97)$$

Note that DDPM posterior is a *stochastic* interpolant, and more importantly, it is self-consistent, which we show in the Lemma below.

Lemma 6. For all $0 \leq s < t \leq 1$, DDPM posterior distribution from t to s as defined in Eq. (97) is a self-consistent Gaussian interpolant between $\mathbf{x}$ and $\mathbf{x}_t$.

Proof. Let $\mathbf{x}_r \sim q_{r|t}(\mathbf{x}_r|\mathbf{x}, \mathbf{x}_t)$ and $\mathbf{x}_s \sim q_{s|r}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_r)$, we show that $\mathbf{x}_s$ follows $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)$.

$$\mathbf{x}_r = \frac{\alpha_t \sigma_r^2}{\alpha_r \sigma_t^2} \mathbf{x}_t + \alpha_r \left(1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}\right) \mathbf{x} + \sigma_r \sqrt{1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}} \epsilon_1 \quad (98)$$

$$\mathbf{x}_s = \frac{\alpha_r \sigma_s^2}{\alpha_s \sigma_r^2} \mathbf{x}_r + \alpha_s \left(1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}\right) \mathbf{x} + \sigma_s \sqrt{1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}} \epsilon_2 \quad (99)$$

where $\epsilon_1, \epsilon_2 \sim \mathcal{N}(0, I)$ are i.i.d. Gaussian noise. Directly expanding

$$\mathbf{x}_s = \frac{\alpha_r \sigma_s^2}{\alpha_s \sigma_r^2} \left[\frac{\alpha_t \sigma_r^2}{\alpha_r \sigma_t^2} \mathbf{x}_t + \alpha_r \left(1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}\right) \mathbf{x} + \sigma_r \sqrt{1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}} \epsilon_1 \right] + \alpha_s \left(1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}\right) \mathbf{x} + \sigma_s \sqrt{1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}} \epsilon_2 \quad (100)$$

$$= \frac{\alpha_t \sigma_s^2}{\alpha_s \sigma_t^2} \mathbf{x}_t + \left[\frac{\alpha_r^2 \sigma_s^2}{\alpha_s \sigma_r^2} \left(1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}\right) + \alpha_s \left(1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}\right) \right] \mathbf{x} + \frac{\alpha_r \sigma_s^2}{\alpha_s \sigma_r^2} \left[\sigma_r \sqrt{1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}} \epsilon_1 \right] + \sigma_s \sqrt{1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_r^2}} \epsilon_2 \quad (101)$$

$$= \frac{\alpha_t \sigma_s^2}{\alpha_s \sigma_t^2} \mathbf{x}_t + \alpha_s \left(1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}\right) \mathbf{x} + \frac{\alpha_r \sigma_s^2}{\alpha_s \sigma_r} \sqrt{1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}} \epsilon_1 + \sigma_s \sqrt{1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}} \epsilon_2 \quad (102)$$

$$\stackrel{(a)}{=} \frac{\alpha_t \sigma_s^2}{\alpha_s \sigma_t^2} \mathbf{x}_t + \alpha_s \left(1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}\right) \mathbf{x} + \sigma_s \sqrt{1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}} \epsilon_3 \quad (103)$$

where (a) is due to the fact that sum of two independent Gaussian variables with variance a^2 and b^2 is also Gaussian with variance $a^2 + b^2$, and $\epsilon_3 \sim \mathcal{N}(0, I)$ is another independent Gaussian noise. We show the calculation of the variance:

$$\begin{aligned} \frac{\alpha_r^2 \sigma_s^4}{\alpha_s^2 \sigma_r^2} \left(1 - \frac{\alpha_t^2 \sigma_r^2}{\alpha_r^2 \sigma_t^2}\right) + \sigma_s^2 \left(1 - \frac{\alpha_r^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}\right) &= \frac{\alpha_r^2 \cancel{\sigma_s^4}}{\cancel{\alpha_s^2} \sigma_r^2} - \frac{\alpha_t^2 \sigma_s^4}{\sigma_t^2 \alpha_s^2} + \sigma_s^2 - \frac{\alpha_r^2 \cancel{\sigma_s^4}}{\cancel{\alpha_s^2} \sigma_t^2} \\ &= \sigma_s^2 \left(1 - \frac{\alpha_t^2 \sigma_s^2}{\alpha_s^2 \sigma_t^2}\right) \end{aligned}$$

This shows $\mathbf{x}_s$ follows $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)$ and completes the proof. $\square$

This shows another possible design of the interpolant that can be used, and diffusion GAN's formulation generally complies with our design of the generative distribution, except that it learns this conditional distribution of $\mathbf{x}$ given $\mathbf{x}_t$ directly while we learn a marginal distribution. When they directly learn the conditional distribution by matching $p_{s|t}^\theta(\mathbf{x}|\mathbf{x}_t)$ with $q_t(\mathbf{x}|\mathbf{x}_t)$, the model is forced to learn $q_t(\mathbf{x}|\mathbf{x}_t)$ and there only exists one minimizer. However, in our case, the model can learn multiple different solutions because we match the marginals instead.

GAN loss and MMD loss. We also want to draw attention to similarity between GAN loss used in [Xiao et al. \(2021\)](#); [Sauer et al. \(2025\)](#) and MMD loss. MMD is an integral probability metric over a set of functions $\mathcal{F}$ in the following form

$$D_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{\mathbf{y} \sim Q(\mathbf{y})} f(\mathbf{y})|$$

where a supremum is taken on this set of functions. This naturally gives rise to an adversarial optimization algorithm if $\mathcal{F}$ is defined as the set of neural networks. However, MMD bypasses this by selecting $\mathcal{F}$ as the RKHS where the optimal f can be analytically found. This eliminates the adversarial objective and gives a stable minimization objective in practice. However, this is not to say that RKHS is the best function set. With the right optimizers and training scheme, the adversarial objective may achieve better empirical performance, but this also makes the algorithm difficult to scale to large datasets.

G.3. Generative Moment Matching Network

It is trivial to check that GMMN is a special parameterization. We fix $t = 1$, and due to boundary condition, $r(s, t) \equiv s = 0$ implies training target $p_{s|r(s,t)}^{\theta^-}(\mathbf{x}_s) = q_s(\mathbf{x}_s)$ is the data distribution. Additionally, $p_{s|t}^\theta(\mathbf{x}_s)$ is a simple pushforward of prior $p(\epsilon)$ through network $g_\theta(\epsilon)$ where drop dependency on t and s since they are constant.